

The first shift theorem enables us to find the inverse LT of a wider variety of fractions which may result from running through the four steps of solving an ODE by LTs. Consider the following example.

Example 6.7.2

Find $y(t)$ when

$$Y(s) = \frac{s}{s^2 - 10s + 29}.$$

can't factorize nicely
⇒ not partial fractions

Solution.

Complete the square:

$$s^2 - 10s + 29 = (s - 5)^2 - 25 + 29 = (s - 5)^2 + 4$$

$$\rightarrow Y(s) = \frac{s}{(s - 5)^2 + 4}$$

← similar to $\mathcal{L}\{e^{5t} \cos 2t\}$
except numerator which needs to be $s - 5$

$$= \frac{s - 5}{(s - 5)^2 + 2^2} + \frac{5}{(s - 5)^2 + 2^2} \times \frac{2}{2}$$

Similar to $\mathcal{L}\{e^{5t} \sin 2t\}$
except numerator, which
needs to be 2.

$$= \frac{s - 5}{(s - 5)^2 + 2^2} + \frac{5}{2} \frac{2}{(s - 5)^2 + 2^2}$$

"Trick" it to get it
in a form we can
transform

$$\therefore y(t) = (e^{5t} \cos 2t + \frac{5}{2} e^{5t} \sin 2t) H(t) \quad (\text{or } t \geq 0)$$



When using the first shift theorem related to sine and cosine functions — results (1) and (2) — one has to be extremely careful about the different role played by s and ω .

Example 6.7.3Find $y(t)$ when

$$Y = \frac{s^2 + 3s + 1}{(s+1)(s+2)^2} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

Solution. $\times (s+1)(s+2)^2: s^2 + 3s + 1 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$

$$Y(s) = \frac{-1}{s+1} + \frac{1}{(s+2)^2}$$

$$= \frac{-1}{s-(-1)} + 2 \frac{1}{s-(-2)} + \frac{1}{(s-(-2))^2}$$

$$\therefore y(t) = -e^{-t} + 2e^{-2t} + e^{-2t}t$$

Not in table, but the unshifted version $\frac{1}{s^2}$ is. $G(s) = \frac{1}{s^2} = \mathcal{L}\{t\}$ $g(t)$
 $\Rightarrow G(s-(-2)) = \frac{1}{(s-(-2))^2} = \mathcal{L}\{e^{-2t}t\}$
 shift by first shift theorem effect of shift

Example 6.7.4

Solve

$$y'' - 2y' + y = 2e^{2t}, \quad y(0) = 0, \quad y'(0) = 1.$$

Solution.

1. Transform

$$\mathcal{L}\{y'' - 2y' + y\} = \mathcal{L}\{2e^{2t}\}$$

 $(y(t) \rightarrow Y(s))$

$$\underbrace{(s^2 Y(s) - \cancel{sy(0)} - \cancel{y'(0)})}_{\mathcal{L}\{y''\}} - 2 \underbrace{(sY(s) - \cancel{y(0)})}_{\mathcal{L}\{y'\}} + Y(s)$$

2. Rearrange

$$(s^2 - 2s + 1)Y(s) = \frac{2}{s-2} + 1$$

$$(s-1)^2 Y(s) = \frac{2}{(s-2)(s-1)^2} + \frac{1}{(s-1)^2}$$

$$\frac{2}{(s-2)(s-1)^2} = \frac{A}{s-2} + \frac{B}{s-1} + \frac{C}{(s-1)^2}$$

$$2 = A(s-1)^2 + B(s-2)(s-1) + C(s-2)$$

$$s=2: \quad 2 = A(1) + 0 + 0 \Rightarrow A = 2$$

$$s=1: \quad 2 = 0 + 0 + C(-1) \Rightarrow C = -2$$

$$s=0: \quad 2 = A(1) + B(2) + C(-2) \Rightarrow B = \frac{1}{2}(2 - A + 2C) \Rightarrow B = -2$$

$$\therefore Y(s) = \frac{2}{s-2} + \frac{-2}{s-1} + \frac{-2}{(s-1)^2} + \frac{1}{(s-1)^2}$$

3. Transform back

 $(Y(s) \rightarrow y(t))$
 $\frac{-1}{(s-1)^2}$ needs 1st shift theorem

$$G(s-1) = \frac{1}{(s-1)^2} \quad a=1$$

$$\Rightarrow G(s) = \frac{1}{s^2}$$

$$\Rightarrow g(t) = t$$

$$\therefore y(t) = 2e^{2t} - 2e^t - e^t t$$



Here is an example with a slightly trickier RHS, but really it just comes down to (a) being able to do partial fractions, and (b) thinking ahead to what you want in terms of entries in the table of Laplace transforms on the formula sheet.

Example 6.7.5

Solve

$$y'' + 4y' + 5y = 40 \sin(3t), \quad y(0) = y'(0) = 0.$$

Solution.



Example 6.7.6

Find the current in the electrical circuit containing, in series, a capacitor (0.025 farads), a resistor (4 ohms), and an inductor (1 henry), when the initial conditions at time $t = 0$ are that the current is 0 and that the charge on the capacitor is Q_0 .

Solution.



Let us continue to extend the range of functions for which we can find the LT.

Example 6.7.7

Show that

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s).$$

Solution.



In general

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

— this is not in the table on the formula sheet, but is quite important here: we use it to build up the range of functions with which we are happy to work, as follows. We know that

$$f(t) = \sin(\omega t) \quad \implies \quad F(s) = \frac{\omega}{s^2 + \omega^2}.$$

So $t \sin(\omega t)$ transforms to

$$-\frac{d}{ds} \left(\frac{\omega}{s^2 + \omega^2} \right) = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}.$$

Similarly

$$\mathcal{L}\{t \cos(\omega t)\} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}.$$

Also

$$\mathcal{L}\{te^{at}\} = \underline{\hspace{4cm}} = \underline{\hspace{4cm}}.$$

More generally

$$\mathcal{L}\{t^n e^{at}\} = \underline{\hspace{4cm}} = \underline{\hspace{4cm}},$$

which can also be proved using the first shift theorem and the fact that

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}.$$

These results together with the first shift theorem can be used to take the inverse Laplace Transform of any proper rational function³. A partial fractions expansion will yield terms like:

$\frac{1}{s^{n+1}}$	with inverse transform	$\frac{t^n}{n!}H(t)$
$\frac{1}{(s-a)^{n+1}}$	"	$\frac{t^n}{n!}e^{at}H(t)$
$\frac{(s-a)}{(s-a)^2 + w^2}$	"	$e^{at} \cos(wt)H(t)$
$\frac{w}{(s-a)^2 + w^2}$	"	$e^{at} \sin(wt)H(t)$
$\frac{2(s-a)w}{((s-a)^2 + w^2)^2}$	"	$e^{at} t \sin(wt)H(t)$

³Proper means that the numerator is a polynomial of lower degree than the denominator.

6.8 SECOND SHIFT (DELAY) THEOREM → IMPORTANT & POWERFUL TOOL

This is the final LT theorem we will look at in this course. It is very important, as it lets us work with piecewise functions, which happen a lot in the real world where we have *sudden changes* (e.g. a circuit being switched on or off, a camera shutter opening and closing, a collision suddenly occurring). The theorem states that if

then

$$\mathcal{L}\{f(t)H(t)\} = F(s)$$

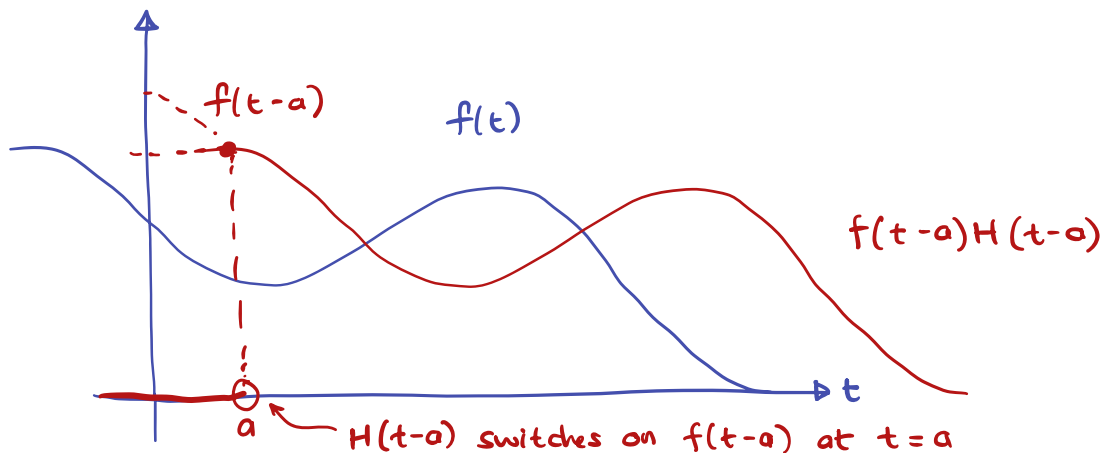
every t must have a "-a" provided $a > 0$. (This is invalid for $a < 0$.)

$$\mathcal{L}\{f(t-a)H(t-a)\} = e^{-as}F(s)$$

Shift right (delay) by a ← constant "a" in the time domain

the effect of the shift

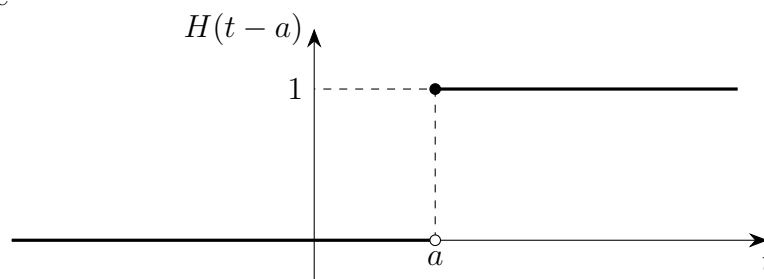
Before we prove this formula, let's consider its meaning. First, sketch any function $f(t)$:



Now draw on the same axes $f(t-a)$ for $a > 0$. We must also have

$$H(t-a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases}$$

which looks like



Therefore $f(t-a)H(t-a)$ is your function shifted to the right by a then switched off for all $t < a$. In other words, it is your function delayed by a (it starts at a rather than 0).